



# TIME SERIES ANALYSIS

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# FORECASTING

$$y_t = \varphi y_{t-1} + e_t$$



$\hat{\varphi}$  (*estimates of  $\varphi$* )



$$\hat{y}_t = \varphi \hat{y}_{t-1}$$



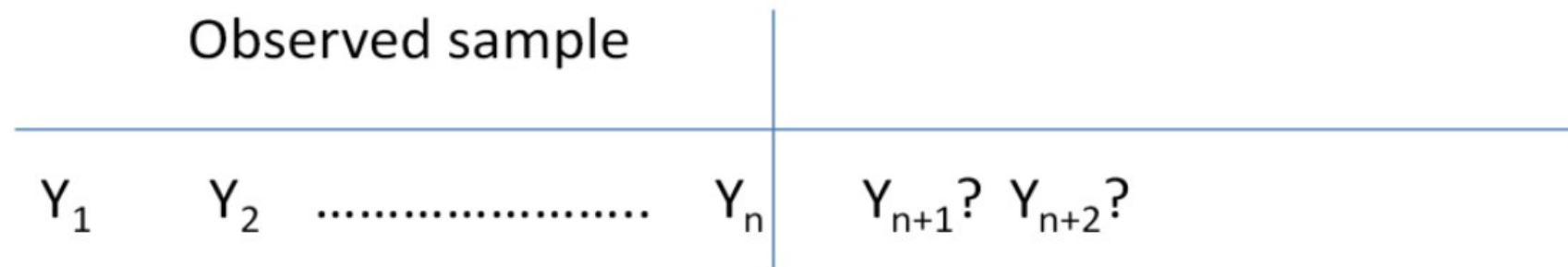
$\hat{y}_{t+1}$  (*forecast*)



# FORECASTING FROM AN ARMA MODEL

## THE MINIMUM MEAN SQUARED ERROR FORECASTS

Observed time series,  $y_1, y_2, \dots, y_n$ .  $n$ : the forecast origin



$\hat{y}_n(1) \rightarrow$  the forecast value of  $y_{n+1}$

$\hat{y}_n(2) \rightarrow$  the forecast value of  $y_{n+2}$

$\hat{y}_n(l) \rightarrow$  the forecast value of  $y_{n+l}$

$\rightarrow$   $l$  step ahead forecast of  $y_{n+1}$

$\rightarrow$  minimum MSE forecast of  $y_{n+1}$



*Forecasting with Stationary AR(1) process:*

$$y_t = \delta + \phi_1 y_{t-1} + u_t, \quad |\phi_1| < 1.$$

We denote  $E[y_{t+h} | \mathcal{F}_t]$  by  $E_t[y_{t+h}]$

$h$ -step forecast is the conditional mean of  $y_{t+h}$ , given  $y_1, \dots, y_t$ , i.e.

$$E_t[y_{t+h}] = E_t[\delta + \phi_1 y_{t+h-1} + u_{t+h}]$$

$$E_t(u_{t+h}) = 0$$

$$= \delta + \phi_1 \underline{E_t[y_{t+h-1}]}$$

Thus

$$\hat{y}_t(h) = \delta + \phi_1 \hat{y}_t(h-1) \quad (13)$$



Solving (13) recursively, we have

$$\text{For } h = 1, \quad \hat{y}_t(1) = \delta + \phi_1 \hat{y}_t(0) = \delta + \phi_1 y_t$$

$$\text{For } h = 2, \quad \hat{y}_t(2) = \delta + \phi_1 \hat{y}_t(1) = \delta + \delta \phi_1 + \phi_1^2 y_t$$

In general

$$\hat{y}_t(h) = \delta \left( \underbrace{1 + \phi_1 + \phi_1^2 + \cdots + \phi_1^{h-1}} \right) + \phi_1^h y_t$$

$$\mu = \frac{\delta}{1 - \phi_1}$$

$$= \frac{\delta(1 - \phi_1^h)}{1 - \phi_1} + \phi_1^h y_t$$

$$= \frac{\delta}{1 - \phi_1} + \phi_1^h \left[ y_t - \frac{\delta}{1 - \phi_1} \right] = \mu + \phi_1^h [y_t - \mu]$$



$\mu = \frac{\delta}{1-\phi_1}$ : Mean of AR(1) process.

As  $h \rightarrow \infty$

$$\hat{y}_t(h) = \mu + \phi_1^h [y_t - \mu] \rightarrow \mu$$

Convergence is monotonic if  $\phi_1 > 0$

Convergence is oscillating if  $\phi_1 < 0$

$$|\phi_1| < 1$$
$$\phi_1^h \rightarrow 0$$



We can write AR(1) process as

$$y_t = \delta + u_t + \phi_1 u_{t-1} + \phi_1^2 u_{t-2} + \dots$$



Forecast error and its variance:

$$f_t(h) = u_{t+h} + \phi_1 u_{t+h-1} + \phi_1^2 u_{t+h-2} + \dots + \phi_1^{h-1} u_{t+1}$$

$$\text{Var}(f_t(h)) = \sigma_u^2 (1 + \phi_1^2 + \dots + \phi_1^{2h-2})$$

$$= \frac{\sigma_u^2 (1 - \phi_1^{2h})}{1 - \phi_1^2}$$

$$f_t(h) = y_{t+h} - \hat{y}_t(h)$$

$\phi_1^{2h} \rightarrow 0$  as  $h \rightarrow \infty$

With increasing  $h$ ,  $\text{Var}(f_t(h))$  increases

$$\text{As } h \rightarrow \infty, \text{Var}(f_t(h)) \rightarrow \sigma_u^2 / (1 - \phi_1^2) = \underline{\text{Var}(y_t)}$$



*Forecasting with Stationary AR(p) process:*

$$y_t = \delta + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + u_t$$

We denote  $E[y_{t+h} | \mathcal{F}_t]$  by  $E_t[y_{t+h}]$ . Then

$$E_t[y_{t+h}] = \delta + \phi_1 E_t[y_{t+h-1}] + \dots + \phi_p E_t[y_{t+h-p}]$$

Here 
$$E_t[y_{t+s}] = \begin{cases} \hat{y}_t(s) & \text{for } s > 0 \\ y_{t+s} & \text{for } s \leq 0 \end{cases}$$

Solving the equation recursively

$$h = 1, \quad \hat{y}_t(1) = \delta + \phi_1 y_t + \dots + \phi_p y_{t+1-p}$$

$$h = 2, \quad \hat{y}_t(2) = \delta + \phi_1 \hat{y}_t(1) + \phi_2 y_t + \dots + \phi_p y_{t+2-p},$$

$$\vdots \quad \hat{y}_t(3) = \delta + \phi_1 \hat{y}_t(2) + \phi_2 \hat{y}_t(1) + \phi_3 y_t + \dots + \phi_p y_{t+3-p}$$

$$E[u_{t+k} | \mathcal{F}_t] = 0$$

$$E_t[y_{t-1}] = E[y_{t-1} | y_1, y_2, \dots, y_t] = y_{t-1}$$



*Forecasting with invertible MA(1) process:*

$$y_t = \mu + u_t - \theta u_{t-1}, |\theta| < 1. \quad (14)$$

Observe that  $E_t(u_{t+1}) = E(u_{t+1}) = 0$ , and  $E_t(u_t) = u_t$ .

Then

$$E_t[y_{t+h}] = \mu + E_t[u_{t+h}] - \theta E_t[u_{t+h-1}] \quad (15) \quad \hat{y}_t^{(2)} = \mu + E_t(u_{t+2}) - \theta E_t(u_{t+1})$$

For  $h=1$ ,

$$\begin{aligned} \hat{y}_t(1) &= \mu + E_t(u_{t+1}) - \theta E_t(u_t) \\ &= \mu - \theta u_t \end{aligned} \quad (16)$$

For all  $h \geq 2$ ,  $\hat{y}_t(h) = \mu$ .



$MA(1)$  process  $\Rightarrow$  ACF  $\rho_k = 0 \forall k \geq 2$ .

Observations up to time  $t$  don't provide any information about  $y_{t+h}$ , for  $h \geq 2$ .

Thus  $\hat{y}_t(h)$  is simply process mean  $\mu$  for  $h \geq 2$ .

$h$ -step ahead prediction errors and their variances:

For  $h = 1$ ,  $f_t(1) = u_{t+1}$ ,

$$\text{Var}[f_t(1)] = \sigma_u^2$$

$\forall h \geq 2$ ,  $f_t(h) = u_{t+h} - \theta_1 u_{t+h-1}$ ;

$$\text{Var}[f_t(h)] = \sigma_u^2(1 + \theta^2) = \text{Var}(y_t).$$

$y_t$   $y_{t+2}$



For obtaining one-step ahead forecast, the unobservable error  $u$  must be expressed in terms of observable variable  $y$ .

For  $s \leq t$ , using (14)  $(y_t = \mu + u_t - \theta u_{t-1})$

and (16)  $(\hat{y}_t(1) = \mu - \theta u_t)$ ,

we obtain

$$y_s - \hat{y}_{s-1}(1) = (\mu + u_s - \theta u_{s-1}) - (\mu - \theta u_{s-1}) = \underline{\underline{u_s}}$$



$$t = 0: \quad \hat{y}_0(1) = \mu - \theta u_0$$

$$\begin{aligned} t = 1: \quad \hat{y}_1(1) &= \mu - \theta u_1 = \mu - \theta(y_1 - \hat{y}_0(1)) \\ &= \mu(1 + \theta) - \theta y_1 - \theta u_0 \end{aligned}$$

$$t = 2: \quad \hat{y}_2(1) = \mu(1 + \theta + \theta^2) - \theta y_2 - \theta^2 y_0 - \theta^3 u_0$$

In general

$$\hat{y}_t(1) = \mu(1 + \theta + \cdots + \theta^t) - \theta y_t - \theta^2 y_{t-1} - \cdots - \theta^t y_1 - \theta^{t+1} u_0$$

Since  $|\theta| < 1$  the impact of  $u_0$  finally disappears.

$$\begin{aligned} |\theta| &< 1 \\ \theta^{t+1} &\rightarrow 0 \end{aligned}$$

Similarly, we can develop forecasts for invertible  $MA(q)$  process.



*Forecasts with ARMA(1,1) process:*

$$y_t = \delta + \phi y_{t-1} + u_t - \theta u_{t-1}, |\phi| < 1, |\theta| < 1, (\phi_1 = \phi, \theta_1 = -\theta)$$

One step ahead forecast:

$$\hat{y}_t(1) = E_t(y_{t+1}) = \delta + \phi y_t - \theta u_t$$

Then

$$\hat{y}_0(1) = \delta + \phi y_0 - \theta u_0 \quad \checkmark$$

$$\hat{y}_1(1) = \delta + \phi y_1 - \theta u_1$$

$$= \delta + \phi y_1 - \theta(y_1 - \hat{y}_0(1))$$

$$= \delta(1 + \theta) + (\phi - \theta)y_1 + \phi\theta y_0 - \theta^2 u_0$$

$$E_t(y_t) = y_t$$

$$E_t(u_{t+1}) = 0$$

$$E_t(u_t) = u_t$$



In general

$$\begin{aligned}\hat{y}_t(1) &= \delta(1 + \theta + \dots + \theta^t) + (\phi - \theta)y_t + \theta(\phi - \theta)y_{t-1} + \\ &\dots + \theta^{t-1}(\phi - \theta)y_1 + \phi\theta^t y_0 - \theta^{t+1}u_0 \\ &= \delta \left( \frac{1 - \theta^{t+1}}{1 - \theta} \right) + (\phi - \theta)y_t + \theta(\phi - \theta)y_{t-1} + \dots + \theta^{t-1}(\phi - \theta)y_1 \\ &\quad + \phi\theta^t y_0 - \theta^{t+1}u_0.\end{aligned}$$

For large  $t$ ,  $\hat{y}_t(1)$  is independent of initial values  $y_0$  and  $u_0$ .

For  $h$ -step ahead forecast,  $h = 2, 3, \dots$ , we proceed recursively as

$$\left. \begin{aligned}\hat{y}_t(2) &= \delta + \phi\hat{y}_t(1), \\ \hat{y}_t(3) &= \delta + \phi\hat{y}_t(2), \\ &\vdots \\ \hat{y}_t(h) &= \delta + \phi\hat{y}_t(h-1).\end{aligned} \right\} \begin{aligned}E_t[y_{t+2}] \\ &= E_t[\delta + \phi y_{t+1} + u_{t+2} - \theta u_{t+1}] \\ &= \delta + \phi \hat{y}_t(1)\end{aligned}$$



*Durbin-Levinson Algorithm:* Prediction equation for a general  $ARMA(p, q)$  process is difficult to evaluate and we use Durbin-Levinson algorithm to obtain solution recursively.

Step 1:  $\phi_{00} = 0, v_1^{(0)} = \gamma_0$

Step 2: For  $n \geq 1$

$$\underline{\underline{\phi_{nn}}} = \frac{\rho_n - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho_{n-k}}{1 - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho_k}$$

For  $n \geq 2, \phi_{nk} = \phi_{n-1,k} - \phi_{nn} \phi_{n-1,n-k}$

Step 3: For  $n \geq 1, v_{n+1}^{(n)} = v_n^{(n-1)} (1 - \phi_{nn}^2)$



*Note: Durbin-Levinson algorithm is also used to obtain PACF*

Recursively, we have

$$n = 1, \phi_{11} = \rho_1$$

$$\hat{y}_1(1) = \rho_1 y_1. \text{ In general } \hat{y}_{n+1}(1) = \rho_1 y_n$$

$$\text{Its MSE } v_2^{(1)} = \gamma_0(1 - \phi_{11}^2)$$

$$n = 2, \phi_{22} = \frac{\rho_2 - \phi_{11}\rho_1}{1 - \phi_{11}\rho_1} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}$$

$$\phi_{21} = \phi_{11} - \phi_{22}\phi_{11} = \rho_1(1 - \phi_{11})$$

$$\hat{y}_{n+1}(2) = \phi_{21}y_n + \phi_{22}y_{n-1}. \text{ Its MSE } v_3^{(2)} = \gamma_0(1 - \phi_{11}^2)(1 - \phi_{22}^2)$$

We continue these steps for  $n = 3, 4, \dots$



For AR(2) process

$$\phi_{11} = \rho_1 = \frac{\phi_1}{1 - \phi_2}$$

$$\phi_{22} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2} = \phi_2$$

$$\phi_{21} = \rho_1(1 - \phi_{22}) = \phi_1$$

$$\phi_{33} = \frac{\rho_3 - \phi_1\rho_2 - \phi_2\rho_1}{1 - \phi_1\rho_1 - \phi_2\rho_2} = 0$$

$$\rho_1(1 - \phi_2) = \phi_1$$

$$\rho_3 = \phi_1\rho_2 + \phi_2\rho_1$$



$$\phi_{31} = \phi_{21} - \phi_{33}\phi_{32} = \phi_1$$

$$\phi_{32} = \phi_{22} - \phi_{33}\phi_{21} = \phi_2$$

$$\phi_{44} = 0$$

Hence, one step ahead predictor is based on two proceeding values:

$$\hat{y}_{n+1}(2) = \phi_1 y_n + \phi_2 y_{n-1}$$

# MEASURING THE FORECAST ACCURACY

| Technique                      | Abbrev | Measures                                                                                               |
|--------------------------------|--------|--------------------------------------------------------------------------------------------------------|
| Mean Squared Error             | MSE    | The average of squared errors over the sample period                                                   |
| Mean Error                     | ME     | The average dollar amount or percentage points by which forecasts differ from outcomes                 |
| Mean Percentage Error          | MPE    | The average of percentage errors by which forecasts differ from outcomes                               |
| Mean Absolute Error            | MAE    | The average of absolute dollar amount or percentage points by which a forecast differs from an outcome |
| Mean Absolute Percentage Error | MAPE   | The average of absolute percentage amount by which forecasts differ from outcomes                      |

# MEASURING THE FORECAST ACCURACY

## 1. Mean Squared Error

The formula used to calculate the mean squared error is:

$$MSE = \frac{1}{n} \sum_{t=1}^n (a_t - f_t)^2$$

## 2. Mean Percentage Error

The formula used to calculate the mean percentage error is:

$$MPE = \frac{1}{n} \sum_{t=1}^n \frac{(a_t - f_t)}{a_t} \times 100$$

## 3. Mean Absolute Error

The formula used to calculate the mean absolute error is:

$$MAE = \frac{1}{n} \sum_{t=1}^n |(a_t - f_t)|$$



#### 4. Mean Absolute Percentage Error

The formula used to calculate the mean absolute percentage error is:

$$MAPE = \frac{1}{n} \sum_{t=1}^n \frac{|(a_t - f_t)|}{a_t} \times 100$$



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## 5. Random Walk -- A Non-stationary Process



## Random walk

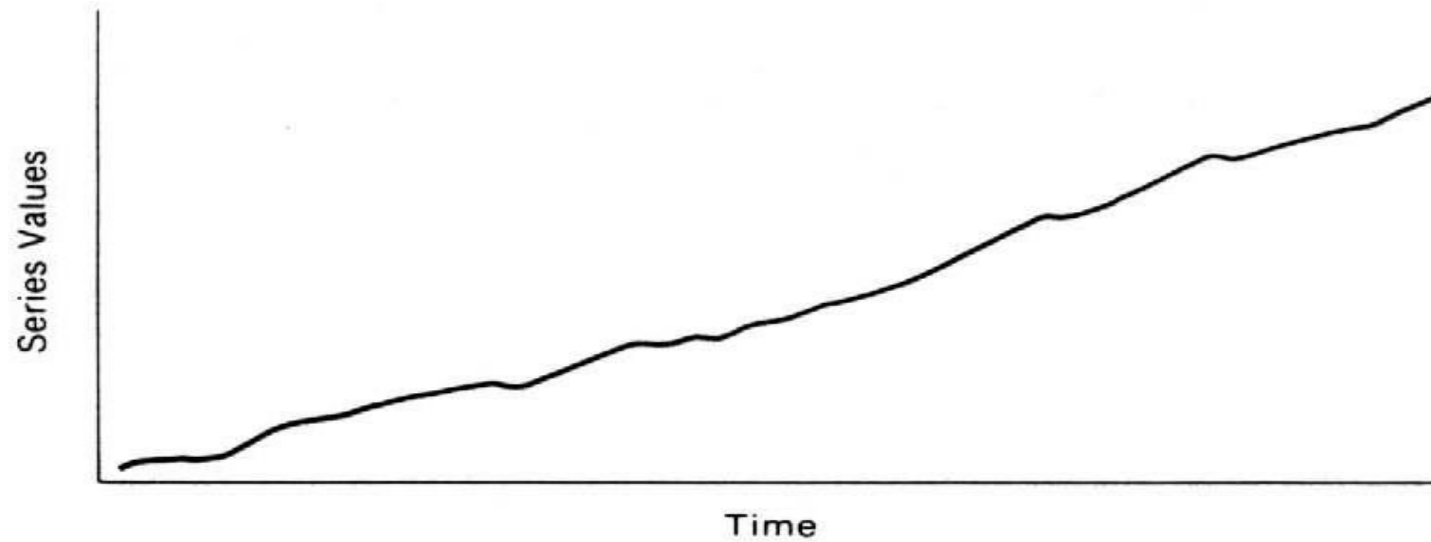
- Start with  $\{y_t\}$  being white noise or purely random
- $\{y_t\}$  is a random walk if

$$y_0 = 0$$

$t$

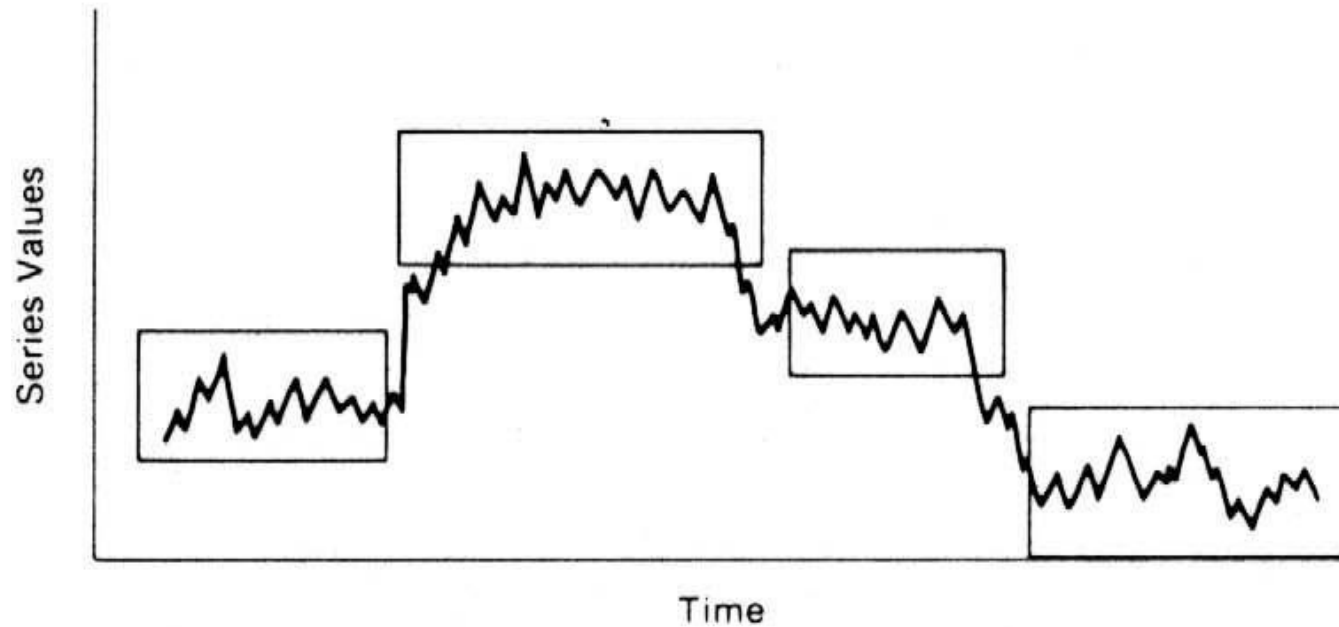
$$y_t = y_t + e_t = \sum_{k=0}^t e_k$$

# Some Other nonstationary series



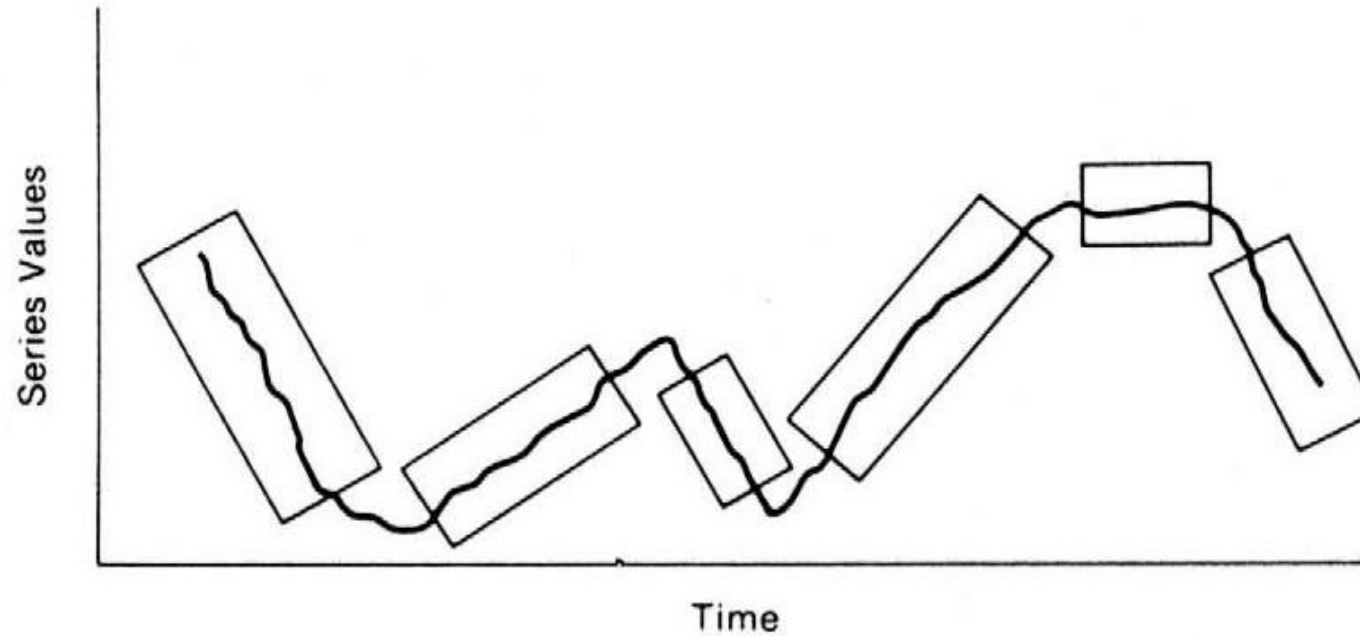
A Nonstationary Series: Overall Trend

## Some nonstationary series (cont )



A Nonstationary Series: Random Changes in Level

## Some nonstationary series (cont.)



A Nonstationary Series: Random Changes in Both Level and Slope

# ARIMA processes

- General autoregressive integrated moving average processes are called ARIMA processes
- When differenced say  $d$  times, the process is an ARMA process
- Call the differenced process  $W_t$ . Then  $W_t$  is an ARMA process and

$$W_t = \Delta^d y_t = (1 - B)^d y_t$$



# ARIMA processes

- Alternatively specify the process as

$$\varphi(B)W_t = \theta(B)e_t$$

Or

$$\varphi(B)(1 - B)^d y_t = \theta(B)e_t$$

- This is an ARIMA process of order (p,d,q)

# ARIMA processes

- The model for  $y_t$  is non-stationary because the AR operator on the left hand side has  $d$  roots on the unit circle
- $d$  is often 1
- Random walk is  $\text{ARIMA}(0,1,0)$
- Can include seasonal terms

# Non-zero mean

- We have assumed that the mean is zero in the ARIMA models
- There are two alternatives
  - mean correct all the  $W_t$  terms in the model
  - incorporate a constant term in the model

# Box-Jenkins Methodology (ARIMA Models)

- I. The Box-Jenkins methodology refers to a set of procedures for identifying and estimating time series models within the class of autoregressive integrated moving average (ARIMA) models.
- II. ARIMA models are regression models that use lagged values of the dependent variable and/or random disturbance term as explanatory variables.
- III. ARIMA models rely heavily on the autocorrelation pattern in the data
- IV. This method applies to both non-seasonal and seasonal data.

# **Box-Jenkins Methods--A five-step iterative procedure**

- I. Stationarity Checking and Differencing
- II. Model Identification
- III. Parameter Estimation
- IV. Diagnostic Checking
- V. Forecasting



# Non-Stationary

- **Not-stationary = Non-stationary**, when distribution (parameters) changes over time.  
Various important examples are:  
**Deterministic trend** and **Stochastic trend**.



# Deterministic Trend (TSP)

$$y_t = \alpha + \beta t + e_t$$

$$E(y_t) = \alpha + \beta t$$

See that mean changes over time.

One can apply OLS to estimate the model parameters.



# Stochastic Trend (DSP)—Unit Root Proces

$$y_t = y_{t-1} + e_t$$

$$y_t = y_0 + e_1 + e_2 + \dots + e_{t-1} + e_t$$

$$E(y_t) = y_0 + E(e_1) + \dots + E(e_t) = y_0$$

$$\begin{aligned}\sigma_{y_t}^2 &= \text{population variance of } (y_0 + e_1 + e_2 + \dots + e_{t-1} + e_t) \\ &= \text{population variance of } (e_1 + e_2 + \dots + e_{t-1} + e_t) \\ &= \sigma_e^2 + \sigma_e^2 + \dots + \sigma_e^2 \\ &= t\sigma_e^2\end{aligned}$$

This process is known as random walk.



# Stochastic Trend

$$y_t = \mu + y_{t-1} + e_t$$

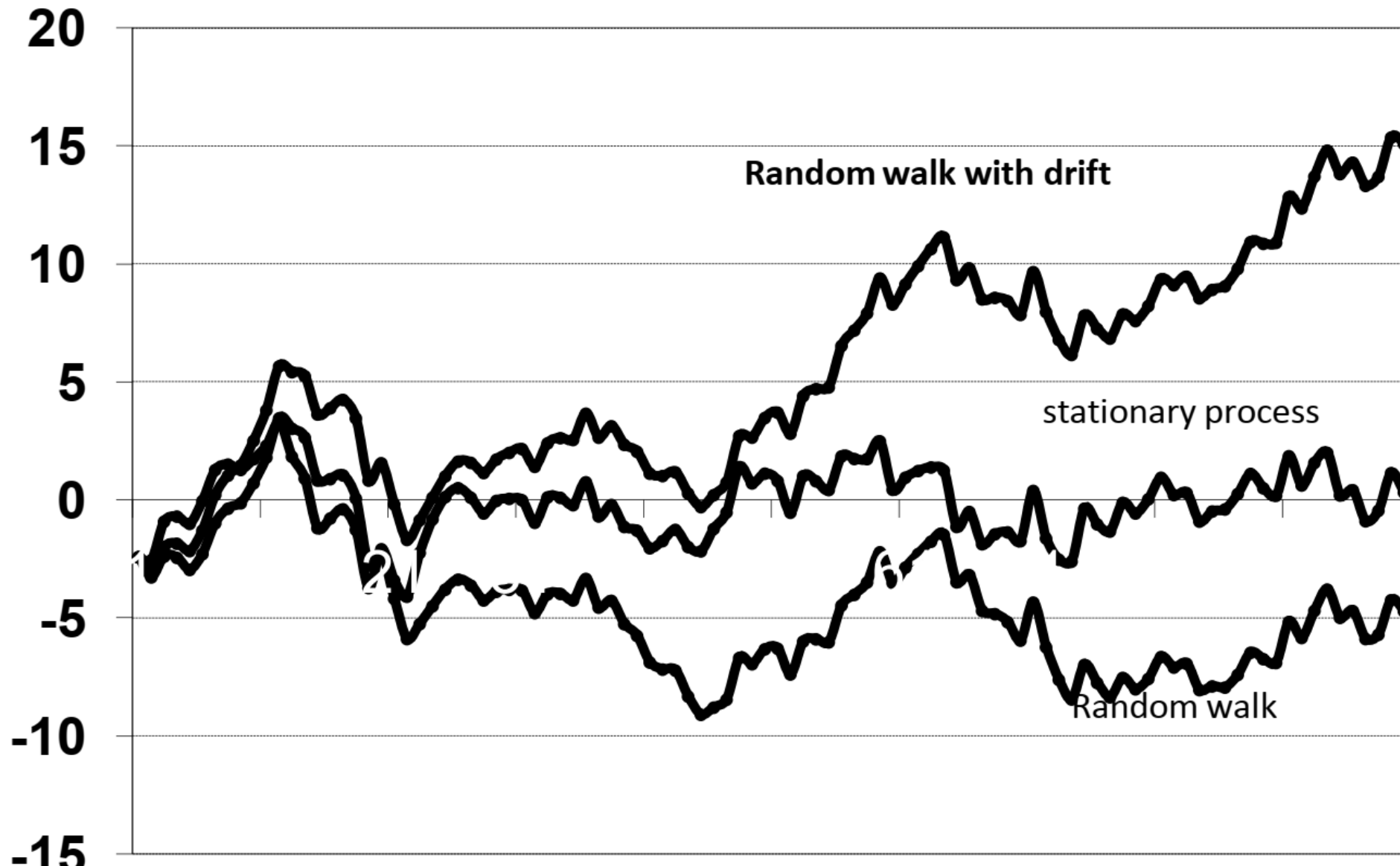
$$y_t = \mu t + y_0 + e_1 + e_2 + \cdots + e_{t-1} + e_t$$

$$E(y_t) = \mu t + y_0$$

$$\sigma_{y_t}^2 = t\sigma_e^2$$

This process is known as random walk with drift.

# Non Stationary Process





# Differencing

- I. Often non-stationary series can be made stationary through differencing.

Examples:

1)  $y_t = y_{t-1} + e_t$  is not stationary, but

$$w_t = y_t - y_{t-1} = e_t \text{ is stationary}$$

2)  $y_t = 1.7y_{t-1} - 0.7y_{t-2} + e_t$  is not stationary, but

$$w_t = y_t - y_{t-1} = 0.7w_{t-1} + e_t \text{ is stationary}$$



# Differencing

- I. Differencing continues until stationarity is achieved.

$$\Delta y_t = y_t - y_{t-1}$$

$$\Delta^2 y_t = \Delta(\Delta y_t) = \Delta(y_t - y_{t-1}) = y_t - 2y_{t-1} + y_{t-2}$$

The differenced series has  $n-1$  values after taking the first-difference,  $n-2$  values after taking the second difference, and so on.

- II. The number of times that the original series must be differenced in order to achieve stationarity is called the order of integration, denoted by  $d$ .
- III. In practice, it is almost never necessary to go beyond second difference, because real data generally involve only first or second level non-stationarity.



# Differencing

- I. The backward shift operator is convenient for describing the process of differencing.

$$\Delta y_t = y_t - y_{t-1} = y_t - B y_t = (1 - B) y_t$$

$$\Delta^2 y_t = y_t - 2y_{t-1} + y_{t-2} = (1 - 2B + B^2) y_t = (1 - B)^2 y_t$$

- II. In general, a  $d$ th-order difference can be written as

$$B^d y_t = (1 - B)^d y_t$$

- III. The backward shift notation is convenient because the terms can be multiplied together to see the combined effect.



- I. If the process is non-stationary then first differences of the series are computed to determine if that operation results in a stationary series.
- II. The process is continued until a stationary time series is found.
- III. This then determines the value of  $d$ .
- IV. Sometimes, transformations, like log or some variance stabilizing transformations are made before ‘Differencing’.



# Step Two: Model Identification

**Determination of the values of  $p$  and  $q$ .**

To determine the value of  $p$  and  $q$  we use the graphical properties of the autocorrelation function and the partial autocorrelation function.

Again recall the following.

**Properties of the ACF and PACF of MA, AR and ARMA Series**

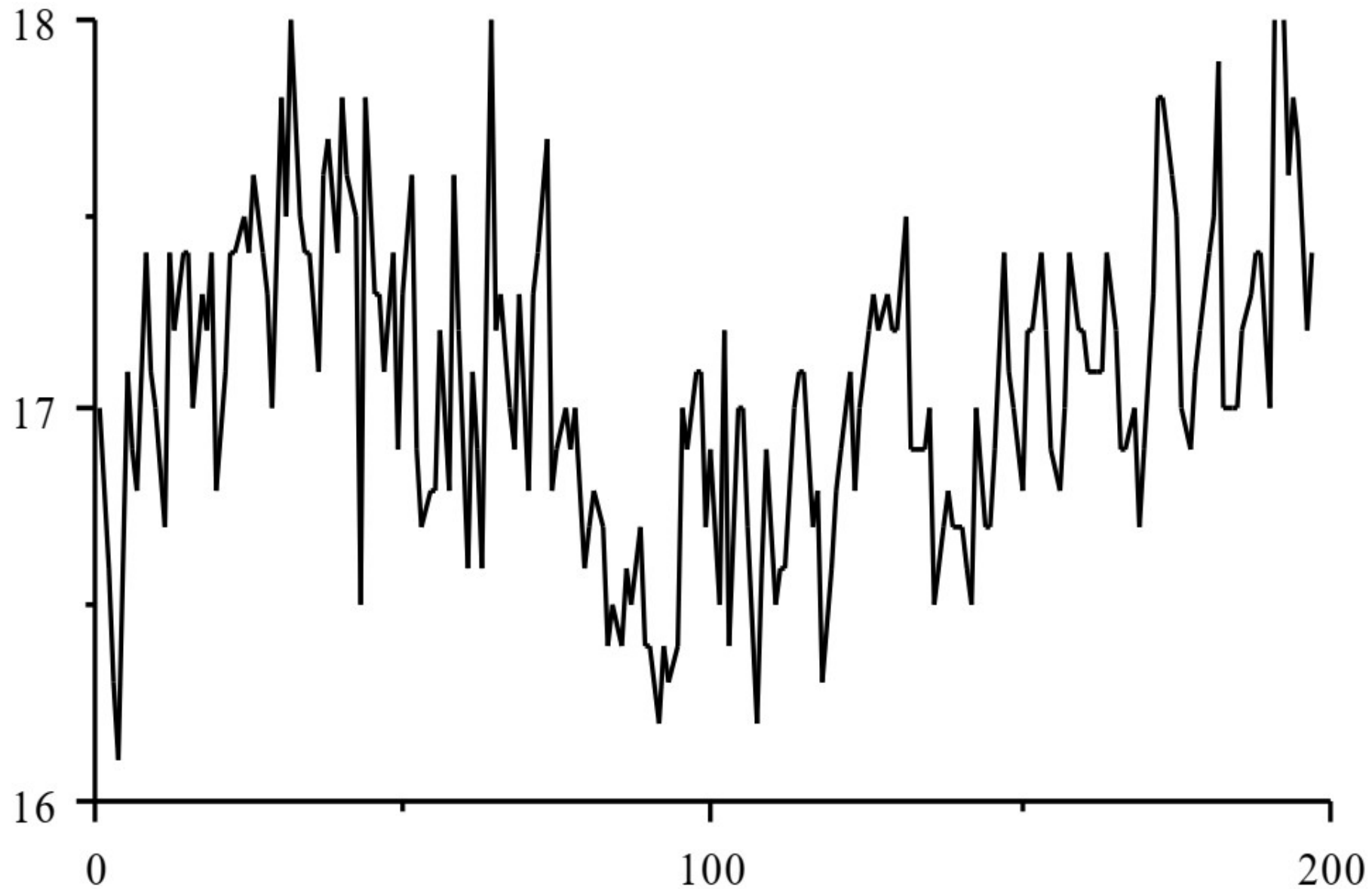
| Process                          | MA( $q$ )                                                                      | AR( $p$ )                                                          | ARMA( $p, q$ )                                                                              |
|----------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Auto-correlation function        | Cuts off                                                                       | Infinite. Tails off.<br>Damped Exponentials<br>and/or Cosine waves | Infinite. Tails off.<br>Damped Exponentials<br>and/or Cosine waves<br>after $q-p$ .         |
| Partial Autocorrelation function | Infinite. Tails off.<br>Dominated by damped<br>Exponentials & Cosine<br>waves. | Cuts off                                                           | Infinite. Tails off.<br>Dominated by damped<br>Exponentials & Cosine waves<br>after $p-q$ . |

**Summary:** To determine  $p$  and  $q$ .  
Use the following table.

|      | MA( $q$ )      | AR( $p$ )      | ARMA( $p, q$ ) |
|------|----------------|----------------|----------------|
| ACF  | Cuts after $q$ | Tails off      | Tails off      |
| PACF | Tails off      | Cuts after $p$ | Tails off      |

**Note:** Usually  $p + q \leq 4$ . There is no harm in over identifying the time series. (Allowing more parameters in the model than necessary. We can always test to determine if the extra parameters are zero.)

## Example A: "Uncontrolled" Concentration, Two-Hourly Readings: Chemical Process





## The data:

|    |      |    |      |     |      |     |      |     |      |
|----|------|----|------|-----|------|-----|------|-----|------|
| 1  | 17.0 | 41 | 17.6 | 81  | 16.8 | 121 | 16.9 | 161 | 17.1 |
| 2  | 16.6 | 42 | 17.5 | 82  | 16.7 | 122 | 17.1 | 162 | 17.1 |
| 3  | 16.3 | 43 | 16.5 | 83  | 16.4 | 123 | 16.8 | 163 | 17.1 |
| 4  | 16.1 | 44 | 17.8 | 84  | 16.5 | 124 | 17.0 | 164 | 17.4 |
| 5  | 17.1 | 45 | 17.3 | 85  | 16.4 | 125 | 17.2 | 165 | 17.2 |
| 6  | 16.9 | 46 | 17.3 | 86  | 16.6 | 126 | 17.3 | 166 | 16.9 |
| 7  | 16.8 | 47 | 17.1 | 87  | 16.5 | 127 | 17.2 | 167 | 16.9 |
| 8  | 17.4 | 48 | 17.4 | 88  | 16.7 | 128 | 17.3 | 168 | 17.0 |
| 9  | 17.1 | 49 | 16.9 | 89  | 16.4 | 129 | 17.2 | 169 | 16.7 |
| 10 | 17.0 | 50 | 17.3 | 90  | 16.4 | 130 | 17.2 | 170 | 16.9 |
| 11 | 16.7 | 51 | 17.6 | 91  | 16.2 | 131 | 17.5 | 171 | 17.3 |
| 12 | 17.4 | 52 | 16.9 | 92  | 16.4 | 132 | 16.9 | 172 | 17.8 |
| 13 | 17.2 | 53 | 16.7 | 93  | 16.3 | 133 | 16.9 | 173 | 17.8 |
| 14 | 17.4 | 54 | 16.8 | 94  | 16.4 | 134 | 16.9 | 174 | 17.6 |
| 15 | 17.4 | 55 | 16.8 | 95  | 17.0 | 135 | 17.0 | 175 | 17.5 |
| 16 | 17.0 | 56 | 17.2 | 96  | 16.9 | 136 | 16.5 | 176 | 17.0 |
| 17 | 17.3 | 57 | 16.8 | 97  | 17.1 | 137 | 16.7 | 177 | 16.9 |
| 18 | 17.2 | 58 | 17.6 | 98  | 17.1 | 138 | 16.8 | 178 | 17.1 |
| 19 | 17.4 | 59 | 17.2 | 99  | 16.7 | 139 | 16.7 | 179 | 17.2 |
| 20 | 16.8 | 60 | 16.6 | 100 | 16.9 | 140 | 16.7 | 180 | 17.4 |
| 21 | 17.1 | 61 | 17.1 | 101 | 16.5 | 141 | 16.6 | 181 | 17.5 |
| 22 | 17.4 | 62 | 16.9 | 102 | 17.2 | 142 | 16.5 | 182 | 17.9 |
| 23 | 17.4 | 63 | 16.6 | 103 | 16.4 | 143 | 17.0 | 183 | 17.0 |
| 24 | 17.5 | 64 | 18.0 | 104 | 17.0 | 144 | 16.7 | 184 | 17.0 |
| 25 | 17.4 | 65 | 17.2 | 105 | 17.0 | 145 | 16.7 | 185 | 17.0 |
| 26 | 17.6 | 66 | 17.3 | 106 | 16.7 | 146 | 16.9 | 186 | 17.2 |
| 27 | 17.4 | 67 | 17.0 | 107 | 16.2 | 147 | 17.4 | 187 | 17.3 |
| 28 | 17.3 | 68 | 16.9 | 108 | 16.6 | 148 | 17.1 | 188 | 17.4 |
| 29 | 17.0 | 69 | 17.3 | 109 | 16.9 | 149 | 17.0 | 189 | 17.4 |
| 30 | 17.8 | 70 | 16.8 | 110 | 16.5 | 150 | 16.8 | 190 | 17.0 |
| 31 | 17.5 | 71 | 17.3 | 111 | 16.6 | 151 | 17.2 | 191 | 18.0 |
| 32 | 18.1 | 72 | 17.4 | 112 | 16.6 | 152 | 17.2 | 192 | 18.2 |
| 33 | 17.5 | 73 | 17.7 | 113 | 17.0 | 153 | 17.4 | 193 | 17.6 |
| 34 | 17.4 | 74 | 16.8 | 114 | 17.1 | 154 | 17.2 | 194 | 17.8 |
| 35 | 17.4 | 75 | 16.9 | 115 | 17.1 | 155 | 16.9 | 195 | 17.7 |
| 36 | 17.1 | 76 | 17.0 | 116 | 16.7 | 156 | 16.8 | 196 | 17.2 |
| 37 | 17.6 | 77 | 16.9 | 117 | 16.8 | 157 | 17.0 | 197 | 17.4 |
| 38 | 17.7 | 78 | 17.0 | 118 | 16.3 | 158 | 17.4 |     |      |
| 39 | 17.4 | 79 | 16.6 | 119 | 16.6 | 159 | 17.2 |     |      |
| 40 | 17.8 | 80 | 16.7 | 120 | 16.8 | 160 | 17.2 |     |      |



## The data:

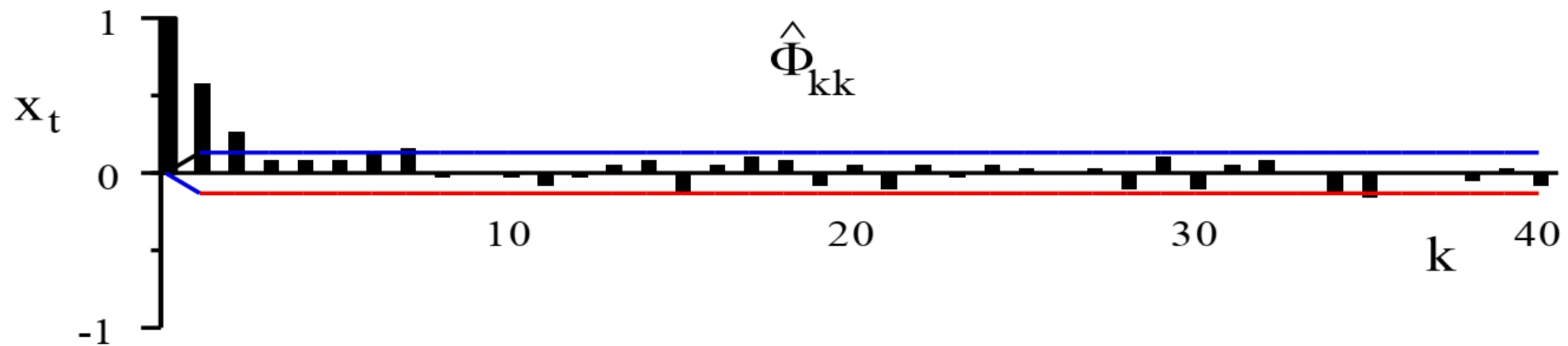
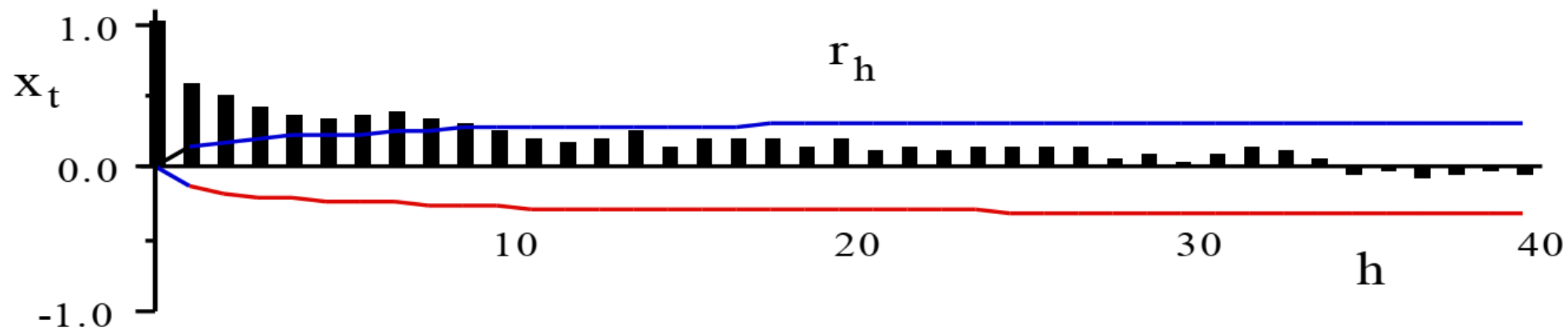
|    |      |    |      |     |      |     |      |     |      |
|----|------|----|------|-----|------|-----|------|-----|------|
| 1  | 17.0 | 41 | 17.6 | 81  | 16.8 | 121 | 16.9 | 161 | 17.1 |
| 2  | 16.6 | 42 | 17.5 | 82  | 16.7 | 122 | 17.1 | 162 | 17.1 |
| 3  | 16.3 | 43 | 16.5 | 83  | 16.4 | 123 | 16.8 | 163 | 17.1 |
| 4  | 16.1 | 44 | 17.8 | 84  | 16.5 | 124 | 17.0 | 164 | 17.4 |
| 5  | 17.1 | 45 | 17.3 | 85  | 16.4 | 125 | 17.2 | 165 | 17.2 |
| 6  | 16.9 | 46 | 17.3 | 86  | 16.6 | 126 | 17.3 | 166 | 16.9 |
| 7  | 16.8 | 47 | 17.1 | 87  | 16.5 | 127 | 17.2 | 167 | 16.9 |
| 8  | 17.4 | 48 | 17.4 | 88  | 16.7 | 128 | 17.3 | 168 | 17.0 |
| 9  | 17.1 | 49 | 16.9 | 89  | 16.4 | 129 | 17.2 | 169 | 16.7 |
| 10 | 17.0 | 50 | 17.3 | 90  | 16.4 | 130 | 17.2 | 170 | 16.9 |
| 11 | 16.7 | 51 | 17.6 | 91  | 16.2 | 131 | 17.5 | 171 | 17.3 |
| 12 | 17.4 | 52 | 16.9 | 92  | 16.4 | 132 | 16.9 | 172 | 17.8 |
| 13 | 17.2 | 53 | 16.7 | 93  | 16.3 | 133 | 16.9 | 173 | 17.8 |
| 14 | 17.4 | 54 | 16.8 | 94  | 16.4 | 134 | 16.9 | 174 | 17.6 |
| 15 | 17.4 | 55 | 16.8 | 95  | 17.0 | 135 | 17.0 | 175 | 17.5 |
| 16 | 17.0 | 56 | 17.2 | 96  | 16.9 | 136 | 16.5 | 176 | 17.0 |
| 17 | 17.3 | 57 | 16.8 | 97  | 17.1 | 137 | 16.7 | 177 | 16.9 |
| 18 | 17.2 | 58 | 17.6 | 98  | 17.1 | 138 | 16.8 | 178 | 17.1 |
| 19 | 17.4 | 59 | 17.2 | 99  | 16.7 | 139 | 16.7 | 179 | 17.2 |
| 20 | 16.8 | 60 | 16.6 | 100 | 16.9 | 140 | 16.7 | 180 | 17.4 |
| 21 | 17.1 | 61 | 17.1 | 101 | 16.5 | 141 | 16.6 | 181 | 17.5 |
| 22 | 17.4 | 62 | 16.9 | 102 | 17.2 | 142 | 16.5 | 182 | 17.9 |
| 23 | 17.4 | 63 | 16.6 | 103 | 16.4 | 143 | 17.0 | 183 | 17.0 |
| 24 | 17.5 | 64 | 18.0 | 104 | 17.0 | 144 | 16.7 | 184 | 17.0 |
| 25 | 17.4 | 65 | 17.2 | 105 | 17.0 | 145 | 16.7 | 185 | 17.0 |
| 26 | 17.6 | 66 | 17.3 | 106 | 16.7 | 146 | 16.9 | 186 | 17.2 |
| 27 | 17.4 | 67 | 17.0 | 107 | 16.2 | 147 | 17.4 | 187 | 17.3 |
| 28 | 17.3 | 68 | 16.9 | 108 | 16.6 | 148 | 17.1 | 188 | 17.4 |
| 29 | 17.0 | 69 | 17.3 | 109 | 16.9 | 149 | 17.0 | 189 | 17.4 |
| 30 | 17.8 | 70 | 16.8 | 110 | 16.5 | 150 | 16.8 | 190 | 17.0 |
| 31 | 17.5 | 71 | 17.3 | 111 | 16.6 | 151 | 17.2 | 191 | 18.0 |
| 32 | 18.1 | 72 | 17.4 | 112 | 16.6 | 152 | 17.2 | 192 | 18.2 |
| 33 | 17.5 | 73 | 17.7 | 113 | 17.0 | 153 | 17.4 | 193 | 17.6 |
| 34 | 17.4 | 74 | 16.8 | 114 | 17.1 | 154 | 17.2 | 194 | 17.8 |
| 35 | 17.4 | 75 | 16.9 | 115 | 17.1 | 155 | 16.9 | 195 | 17.7 |
| 36 | 17.1 | 76 | 17.0 | 116 | 16.7 | 156 | 16.8 | 196 | 17.2 |
| 37 | 17.6 | 77 | 16.9 | 117 | 16.8 | 157 | 17.0 | 197 | 17.4 |
| 38 | 17.7 | 78 | 17.0 | 118 | 16.3 | 158 | 17.4 |     |      |
| 39 | 17.4 | 79 | 16.6 | 119 | 16.6 | 159 | 17.2 |     |      |
| 40 | 17.8 | 80 | 16.7 | 120 | 16.8 | 160 | 17.2 |     |      |



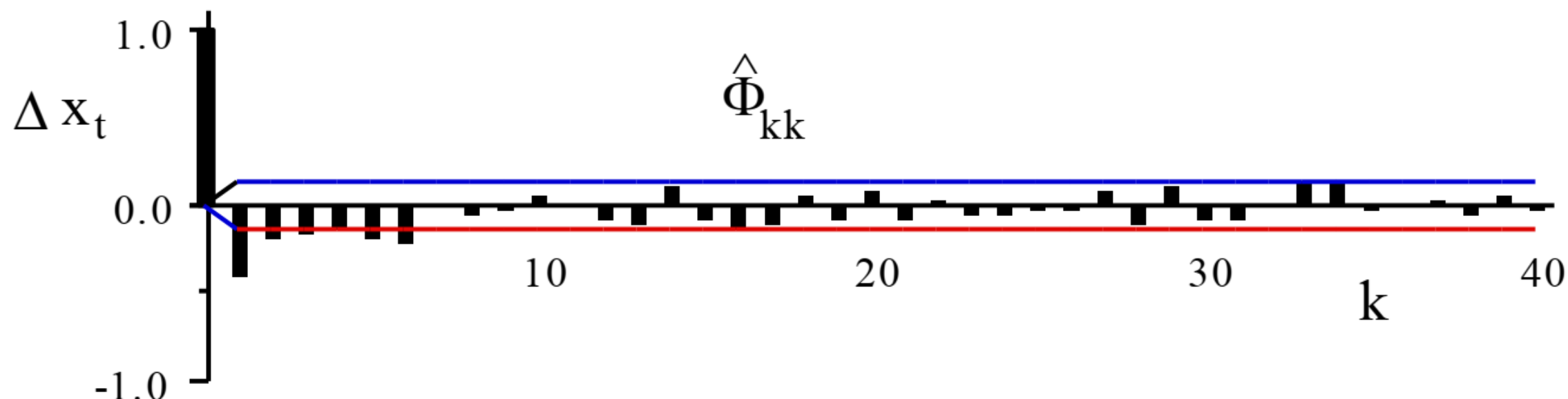
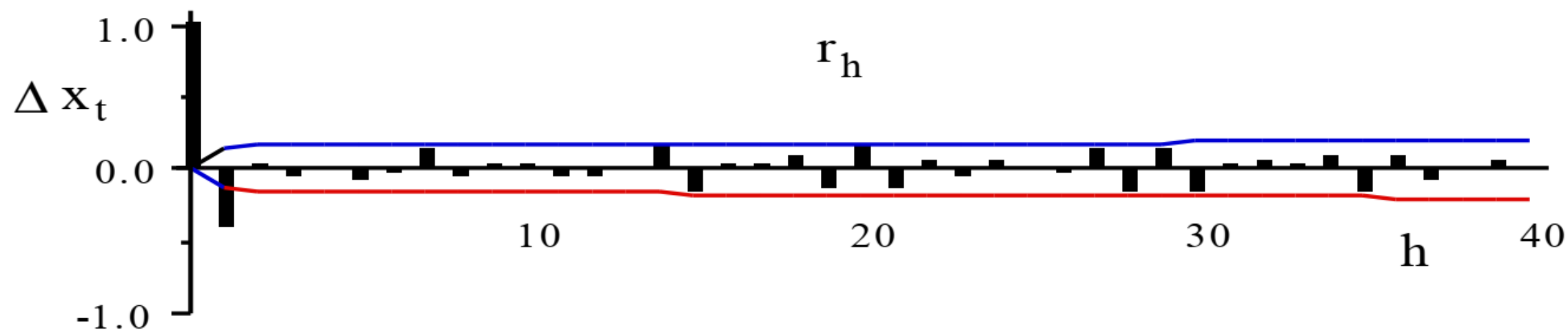
par **Summary Statistics**

| d | N   | Me an  | Std. Dev. |
|---|-----|--------|-----------|
| 0 | 197 | 17.062 | 0.398     |
| 1 | 196 | 0.002  | 0.369     |
| 2 | 195 | 0.003  | 0.622     |

# ACF and PACF for $X_t$ , $\Delta X_t$ and $\Delta^2 X_t$ Chemical concentration DATA

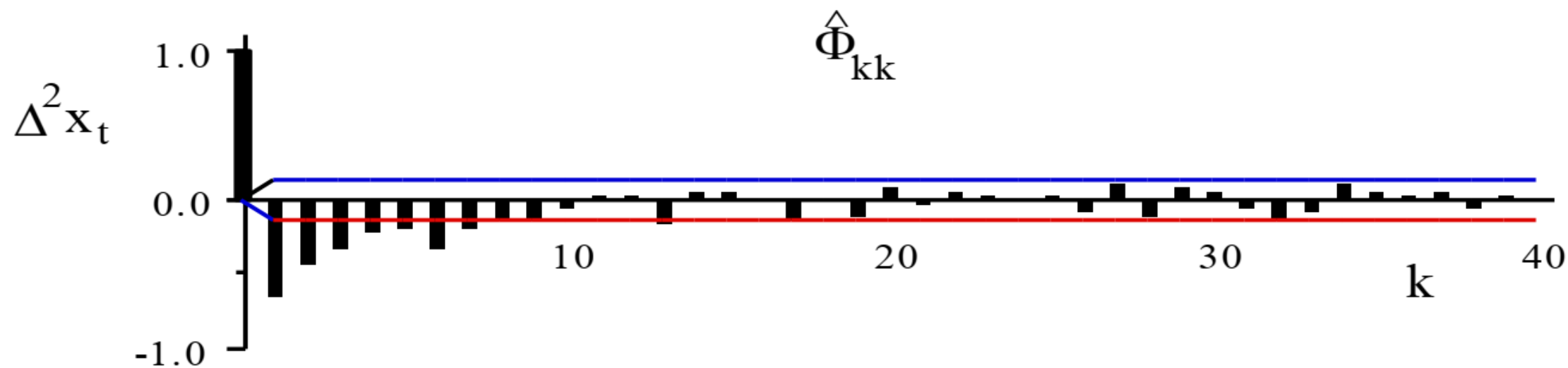
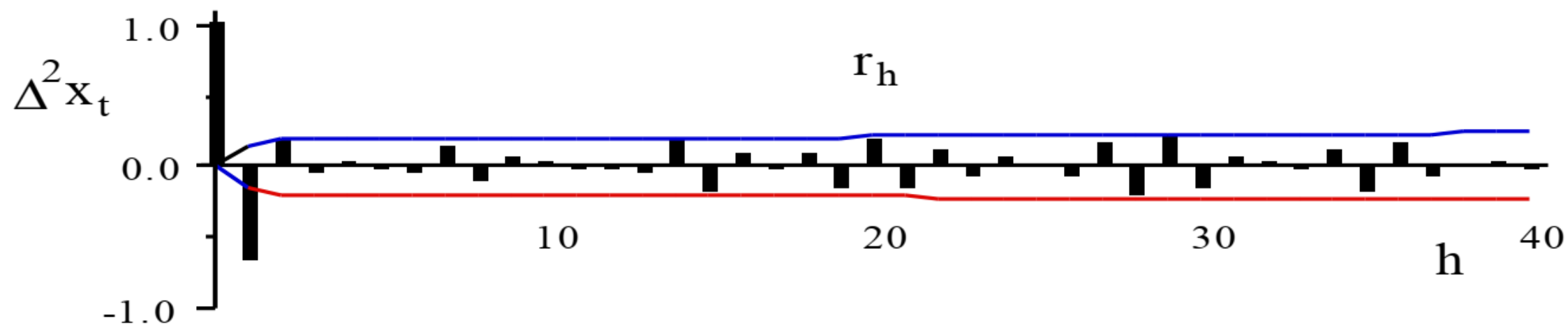


# ACF and PACF for $X_t$ , $\Delta X_t$ and $\Delta^2 X_t$ Chemical concentration DATA





# ACF and PACF for $y_t, \Delta y_t$ and $\Delta^2 y_t$ Chemical concentration DATA





# Possible Identifications

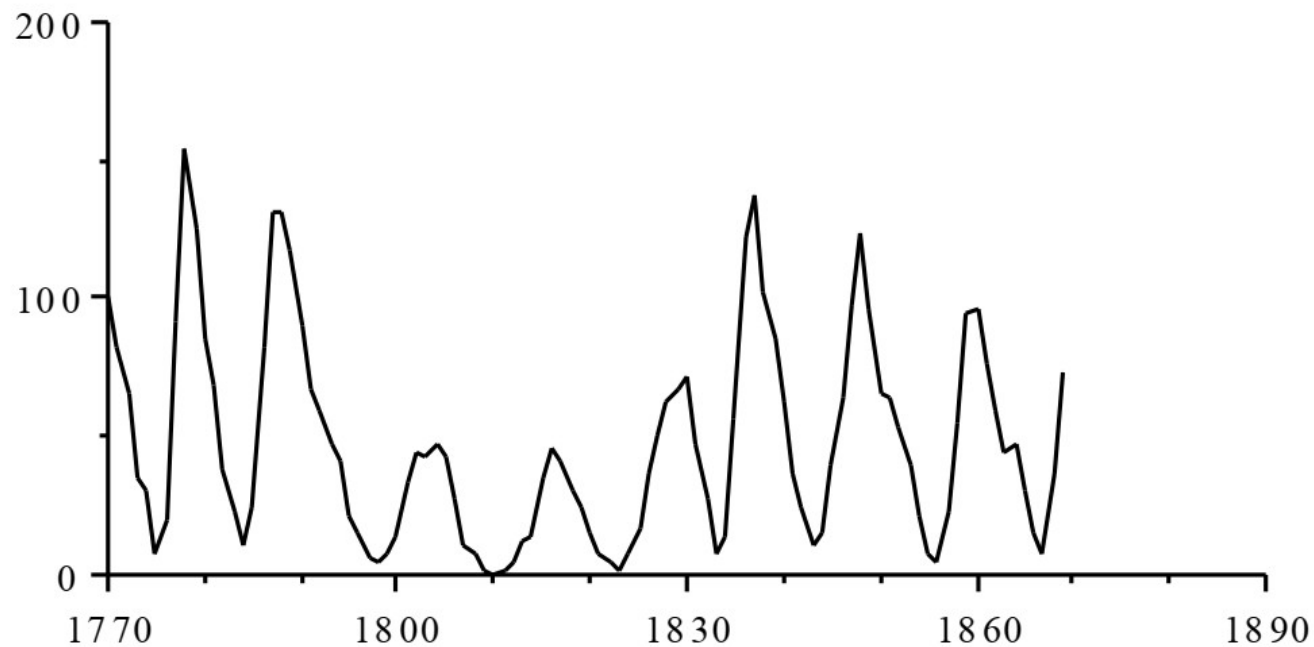
$$1. d = 0, p = 1, q = 1$$

$$2. d = 1, p = 0, q = 1$$



## Sunspot Data:

### Example B: Annual Sunspot Numbers (1790-1869)



### Summary Statistics for the Sunspot Data

| d | N   | mean   | Std. Dev. |
|---|-----|--------|-----------|
| 0 | 100 | 46.950 | 37.186    |
| 1 | 99  | -0.273 | 22.440    |
| 2 | 98  | 0.571  | 20.198    |

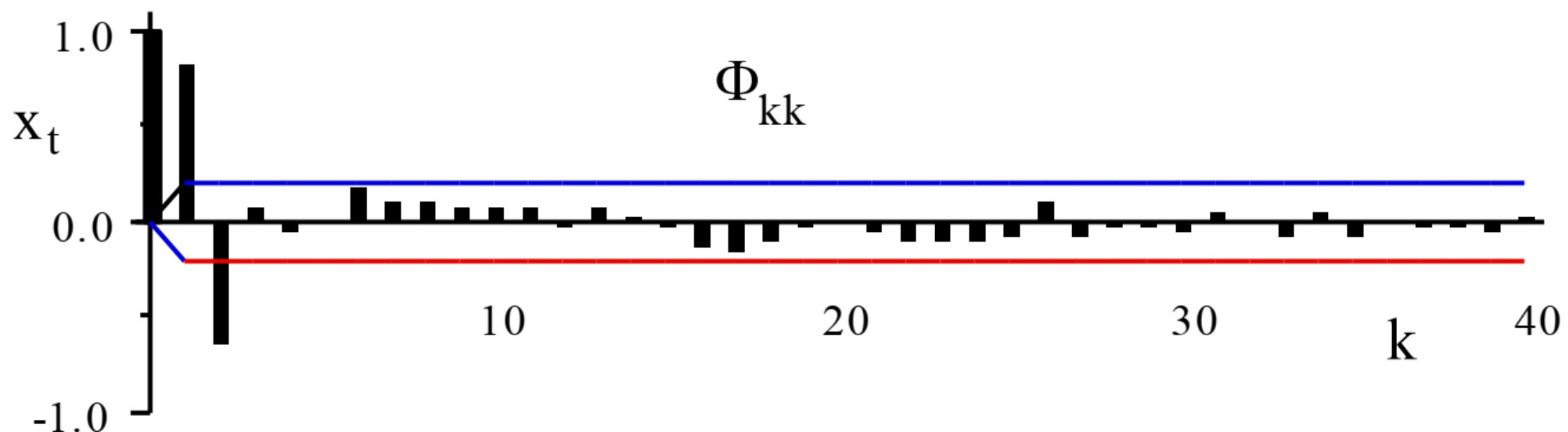
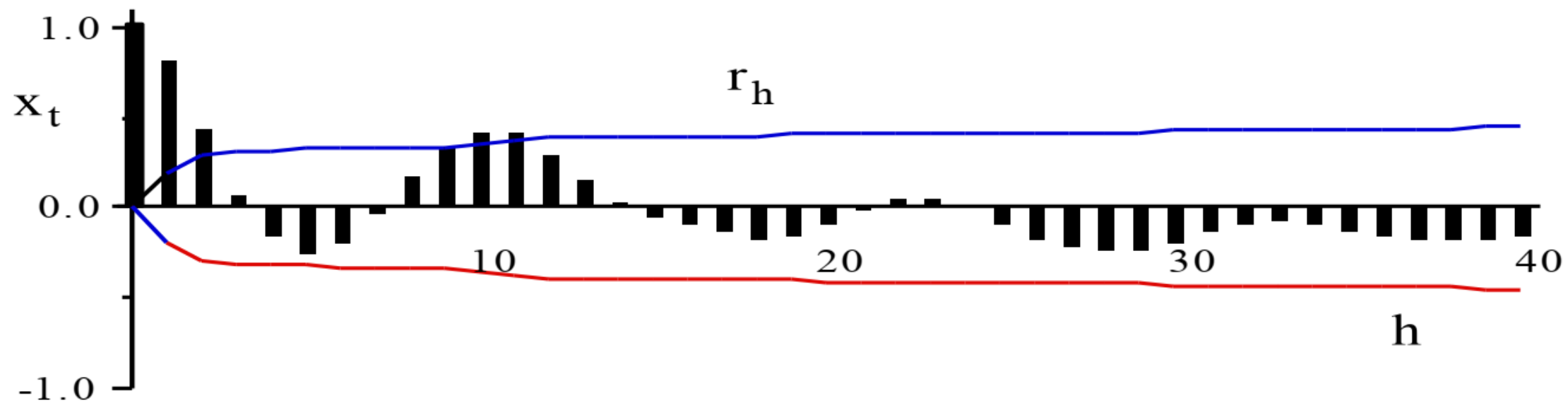


## Example B: Sunspot Numbers: Yearly

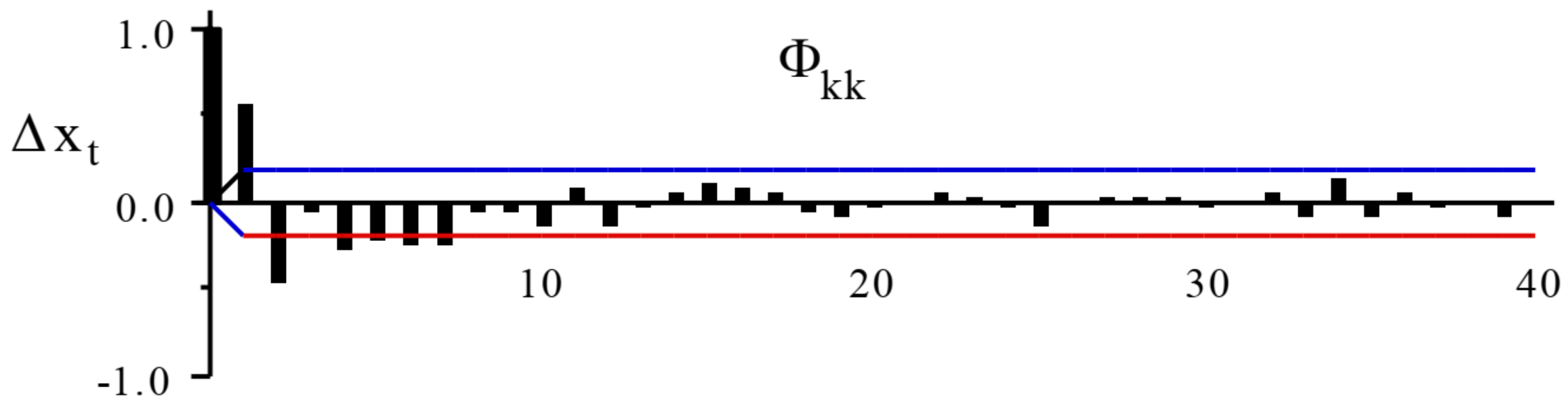
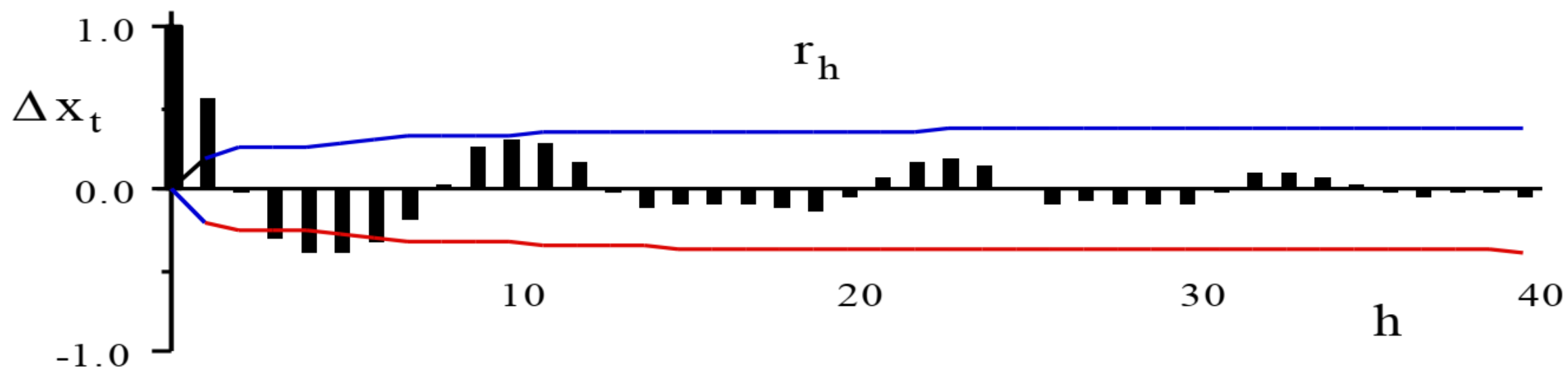
The  
Data:

|      |     |      |    |      |     |      |     |
|------|-----|------|----|------|-----|------|-----|
| 1770 | 101 | 1795 | 21 | 1820 | 16  | 1845 | 40  |
| 1771 | 82  | 1796 | 16 | 1821 | 7   | 1846 | 64  |
| 1772 | 66  | 1797 | 6  | 1822 | 4   | 1847 | 98  |
| 1773 | 35  | 1798 | 4  | 1823 | 2   | 1848 | 124 |
| 1774 | 31  | 1799 | 7  | 1824 | 8   | 1849 | 96  |
| 1775 | 7   | 1800 | 14 | 1825 | 17  | 1850 | 66  |
| 1776 | 20  | 1801 | 34 | 1826 | 36  | 1851 | 64  |
| 1777 | 92  | 1802 | 45 | 1827 | 50  | 1852 | 54  |
| 1778 | 154 | 1803 | 43 | 1828 | 62  | 1853 | 39  |
| 1779 | 125 | 1804 | 48 | 1829 | 67  | 1854 | 21  |
| 1780 | 85  | 1805 | 42 | 1830 | 71  | 1855 | 7   |
| 1781 | 68  | 1806 | 28 | 1831 | 48  | 1856 | 4   |
| 1782 | 38  | 1807 | 10 | 1832 | 28  | 1857 | 23  |
| 1783 | 23  | 1808 | 8  | 1833 | 8   | 1858 | 55  |
| 1784 | 10  | 1809 | 2  | 1834 | 13  | 1859 | 94  |
| 1785 | 24  | 1810 | 0  | 1835 | 57  | 1860 | 96  |
| 1786 | 83  | 1811 | 1  | 1836 | 122 | 1861 | 77  |
| 1787 | 132 | 1812 | 5  | 1837 | 138 | 1862 | 59  |
| 1788 | 131 | 1813 | 12 | 1838 | 103 | 1863 | 44  |
| 1789 | 118 | 1814 | 14 | 1839 | 86  | 1864 | 47  |
| 1790 | 90  | 1815 | 35 | 1840 | 63  | 1865 | 30  |
| 1791 | 67  | 1816 | 46 | 1841 | 37  | 1866 | 16  |
| 1792 | 60  | 1817 | 41 | 1842 | 24  | 1867 | 7   |
| 1793 | 47  | 1818 | 30 | 1843 | 11  | 1868 | 37  |
| 1794 | 41  | 1819 | 24 | 1844 | 15  | 1869 | 74  |

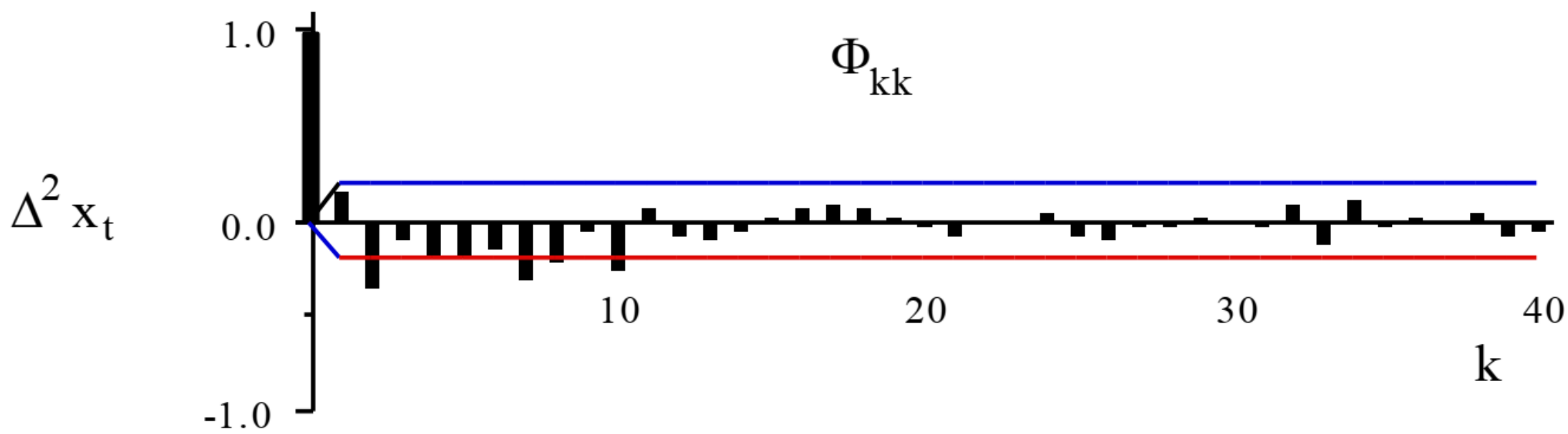
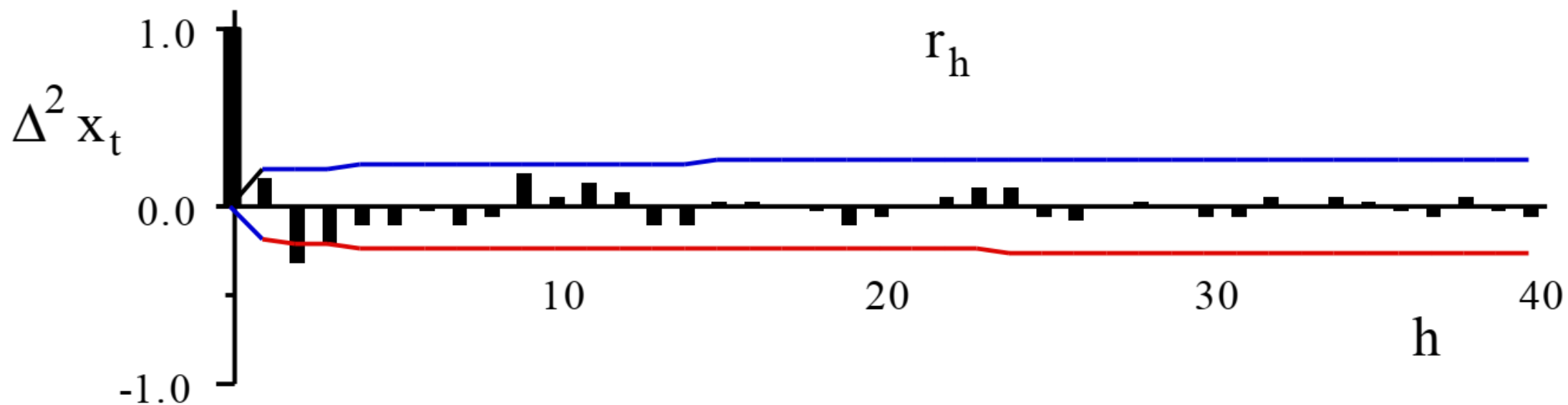
# ACF and PACF for $y_t$ , $\Delta y_t$ and $\Delta^2 y_t$ Sunspot Data



# ACF and PACF for $y_t$ , $\Delta y_t$ and $\Delta^2 y_t$ Sunspot Data



# ACF and PACF for $y_t$ , $\Delta y_t$ and $\Delta^2 y_t$ Sunspot Data





# Possible Identification

$$1. d = 0, p = 2, q = 0$$



# Step Three: Parameter Estimation



# Example (Chemical Concentration Data)

## Possible Identifications

$$1. d = 0, p = 1, q = 1$$

$$2. d = 1, p = 0, q = 1$$

$$y_t = 0.87 y_{t-1} + e_t - 0.48 e_t + 2.25$$



# Step Four: Diagnostic Checking



# Diagnostic Checking

- Often it is not straightforward to determine a single model that most adequately represents the data generating process, and it is not uncommon to estimate several models at the initial stage. The model that is finally chosen is the one considered best based on a set of diagnostic checking criteria. These criteria include
  - (1) t-tests for coefficient significance
  - (2) residual analysis
  - (3) model selection criteria



# Residual Analysis

- If an  $ARMA(p,q)$  model is an adequate representation of the data generating process, then the residuals should be uncorrelated.